

PRACTICE SET B

Operational Research & Optimisation

Revision Questions

- LP Formulation & Graphical Method
- Sensitivity Analysis
- Integer & Goal Programming
- Markov Chains & Steady State
- Network Models: Shortest Route, Spanning Tree, Max Flow
- Transportation & Assignment

Q1**3 pts**

LP FORMULATION

A bakery makes cakes (£8 profit) and pastries (£5 profit). Each cake needs 2 kg flour and 1 h oven time. Each pastry needs 1 kg flour and 0.5 h oven time. There are 80 kg flour and 30 h oven time available. Which of these is the CORRECT objective function?

AMaximise $2C + P$ (resource totals)**B**Maximise $8C + 5P$ (profit)**C**Minimise $8C + 5P$ (profit)**D**Maximise $80C + 30P$ (resource limits)

Correct Answer:

B — Maximise $8C + 5P$ **Explanation**

The objective is to maximise total profit. Profit per cake = £8 and per pastry = £5, so the objective function is Maximise $8C + 5P$. Resources (flour, oven time) form the constraints, not the objective.

Q2

2 pts

LP — GRAPHICAL METHOD

When solving a two-variable LP graphically, the set of all points that satisfy every constraint simultaneously is called the:

A Optimal point

B Feasible region

C Objective plane

D Simplex tableau

Correct Answer:

B — Feasible region**Explanation**

The feasible region is the set of all points (x_1, x_2) that satisfy all constraints including non-negativity. The optimal solution must lie within or on the boundary of this region (at a corner point for LP).

Q3**4 pts**

SENSITIVITY ANALYSIS

A production LP has this sensitivity output:

Labour constraint: Shadow Price = 8, RHS = 120 h, Allowable Increase = 15, Allowable Decrease = 20

A manager can buy 10 extra hours of labour. What is the maximum extra profit this could generate?

A£80 (shadow price $\times$ 10 hrs)**B**

£120 (RHS value)

C

£8 (just the shadow price)

D

Cannot be determined

Correct Answer:

A — £80**Explanation**

Adding 10 hours is within the allowable increase ($10 \leq 15$), so the shadow price is valid over this range. Extra profit = Shadow Price $\times$ extra units = $8 \times 10 = £80$.

Q4

4 pts

SENSITIVITY ANALYSIS

A sensitivity report shows:

Product Z: Objective Coefficient = 50

Allowable Increase = 12

Allowable Decrease = 18

$\times$ 35

$$32 \leq C \leq 72$$

$$50 + 12 = 72$$

$$50 - 18 = 32$$

The market price of Z drops, changing its coefficient to 35. Does the current optimal basis remain optimal?

A Yes — decrease of 15 is within the allowable decrease of 18

B No — the coefficient fell below the allowable range

C Can't tell without re-solving the LP

D Yes — any price drop keeps the basis optimal

$$50 - 35 = 15 \downarrow$$

Correct Answer:

A — Yes, basis remains optimal**Explanation**

The coefficient decreases by $50 - 35 = 15$. The allowable decrease is 18. Since $15 \leq 18$, the change is within the sensitivity range, so the current optimal basis remains valid without re-solving.

Q5**3 pts**

INTEGER PROGRAMMING

A 0-1 IP model includes the constraint:

$$x_1 + x_2 \leq 2$$

What does this constraint mean?

A

Exactly two projects must be selected

B

At most two projects may be selected

C

All two projects must be selected

D

Projects 1 and 2 cannot both be selected

Correct Answer:

B — At most two projects may be selected**Explanation**

Since each $x_i \in \{0,1\}$, the constraint $x_1 + x_2 \leq 2$ limits the total number selected to two or fewer. It does not force exactly two to be chosen — selecting zero or one is also feasible.

Q6**3 pts**

INTEGER PROGRAMMING

In a 0-1 IP, you want to ensure that project A and project B CANNOT both be selected simultaneously. Which constraint achieves this?

A

$$x_A + x_B \geq 1$$

B

$$x_A + x_B = 1$$

C

$$x_A + x_B \leq 1$$

D

$$x_A - x_B = 0$$

Correct Answer:

$$C — xA + xB \leq 1$$

Explanation

$x_A + x_B \leq 1$ means at most one of A or B can equal 1. This enforces mutual exclusivity. Note: $x_A + x_B = 1$ would force exactly one to be chosen (which is a stronger requirement).

A hospital sets a goal of treating at least 200 patients per week. Actual throughput is 185 patients. Which deviational variables apply and what are their values?

A $d^+ = 15, d^- = 0$ (over-achieved by 15)

B $d^+ = 0, d^- = 15$ (under-achieved by 15)

C $d^+ = 185, d^- = 200$

D $d^+ = 200, d^- = 185$

$$d^- = 15 \quad d^+ = 0 \quad 200 - 185 = 15$$

Correct Answer:

B — $d^+ = 0$, $d^- = 15$

Explanation

The hospital fell short by $200 - 185 = 15$ patients. This is under-achievement, so $d^- = 15$ and $d^+ = 0$. In goal programming, d^- measures how far below the target the solution is, while d^+ measures over-achievement.

Q8

3 pts

GOAL PROGRAMMING

In a goal programming model, Goal 1 is cost, and Goal 2 is quality. The optimal solution gives $d_1^- = 0$ (cost met), $d_1^+ = 0$, $d_2^- = 4$ (quality missed by 4), $d_2^+ = 0$. What is the objective function value?

A

4

B

12

C

0

D

16

↓ $Z = d_1^+ + d_2^-$
 $Z = 0 + 4$

Correct Answer:

A — 4**Explanation**

The objective is $\text{Min } d_1^- + d_2^-$ (minimise weighted under-deviations). Substituting: $1(0) + 1(4) = 4$. Only the quality goal is missed, contributing 4 to the objective.

Q9

2 pts

MARKOV CHAINS

A Markov chain has three states: Sunny (S), Cloudy (C), Rainy (R). The row for state C in the transition matrix is $[0.3, 0.5, 0.2]$. What does the entry 0.5 represent?

A

 $P(C \text{ today} \mid S \text{ yesterday})$

B

 $P(C \text{ tomorrow} \mid C \text{ today})$

C

 $P(S \text{ tomorrow} \mid C \text{ today})$

D

 $P(R \text{ today} \mid C \text{ yesterday})$

Correct Answer:

B — $P(\text{C tomorrow} \mid \text{C today}) = 0.5$ **Explanation**

In a transition matrix, the entry in row C, column C is the probability of staying in state C: $P(\text{Cloudy tomorrow} \mid \text{Cloudy today}) = 0.5$. Rows index the current state; columns index the next state.

Q10**5 pts**

MARKOV CHAINS

A machine is either Working (W) or Broken (B). Transition probabilities:

$$P(W \rightarrow W) = 0.90, \quad P(W \rightarrow B) = 0.10$$

$$P(B \rightarrow W) = 0.40, \quad P(B \rightarrow B) = 0.60$$

The machine is Working today. What is the probability it is Working in 2 days?

A

0.810

B

0.850

C

0.854

D

0.900

Correct Answer:

B — 0.850**Explanation**

Day 1 from W: $P(W) = 0.90$, $P(B) = 0.10$

Day 2: $P(W) = P(W \text{ day1}) \times 0.90 + P(B \text{ day1}) \times 0.40$

$$= 0.90 \times 0.90 + 0.10 \times 0.40$$

$$= 0.810 + 0.040 = 0.850$$

Wait — let me recalculate carefully:

$$= 0.81 + 0.04 = 0.85$$

The answer is 0.850.

Q11**5 pts**

MARKOV CHAINS — STEADY STATE

A supermarket's customers choose between Brand X and Brand Y. Each week, 35% of Brand X customers switch to Y, and 20% of Brand Y customers switch to X. If there are currently 500 customers each, what is the long-run market share of Brand Y?

A

63.6%

B

36.4%

C

35%

D

65%

Correct Answer:

A — 63.6%**Explanation**

Steady state: $\pi_X \times 0.35 = \pi_Y \times 0.20$ and $\pi_X + \pi_Y = 1$

$$0.35(1 - \pi_Y) = 0.20 \cdot \pi_Y$$

$$0.35 = 0.55 \cdot \pi_Y$$

$$\pi_Y = 0.35 / 0.55 \approx 0.636 = 63.6\%$$

Brand X holds the remaining 36.4%.

Q12

2 pts

NETWORK MODELS

In a network flow problem, the node where all flow originates (net outflow = 1) is called the:

A

Sink

B

Transshipment node

C

Source

D

Terminal

Correct Answer:

C — Source**Explanation**

The source is the origin node with a net outflow of +1 (flow leaves but does not enter). The sink is the destination with net inflow of +1. Intermediate nodes have zero net flow (flow conservation).

Q13**5 pts**

SHORTEST ROUTE

Find the shortest path from A to E.

$A \rightarrow B = 4$, $A \rightarrow C = 2$, $B \rightarrow D = 5$, $C \rightarrow B = 1$, $C \rightarrow D = 8$, $B \rightarrow E = 3$, $D \rightarrow E = 2$

A $A \rightarrow B \rightarrow E = 7$ **B** $A \rightarrow C \rightarrow B \rightarrow E = 6$ **C** $A \rightarrow C \rightarrow B \rightarrow D \rightarrow E = 9$ **D** $A \rightarrow B \rightarrow D \rightarrow E = 11$

Correct Answer:

B — $A \rightarrow C \rightarrow B \rightarrow E = 6$ **Explanation**

Path evaluation:

- $A \rightarrow B \rightarrow E = 4 + 3 = 7$
- $A \rightarrow C \rightarrow B \rightarrow E = 2 + 1 + 3 = 6$ ✓ shortest
- $A \rightarrow C \rightarrow B \rightarrow D \rightarrow E = 2 + 1 + 5 + 2 = 10$
- $A \rightarrow C \rightarrow D \rightarrow E = 2 + 8 + 2 = 12$

Shortest = $A \rightarrow C \rightarrow B \rightarrow E$ with total distance 6.

When building a minimal spanning tree using Prim's algorithm starting from node 1, which principle guides arc selection at each step?

A Add the longest arc connected to any visited node

B Add the shortest arc that connects a visited node to an unvisited node without forming a cycle

C Add any arc that connects two unvisited nodes

D Add arcs in alphabetical order of their endpoints

Correct Answer:

B — Shortest arc from visited to unvisited (no cycle)**Explanation**

Prim's algorithm grows the spanning tree one arc at a time: at each step, select the cheapest arc connecting the current tree to any node not yet included, provided it does not create a cycle. This greedy rule guarantees a minimum spanning tree.

Q15**3 pts**

MAXIMAL FLOW

In a maximal-flow problem, an arc has forward capacity 8 and current flow 5. What is the residual (spare) capacity in the forward direction?

A

13

B

5

C

3

D

8

Correct Answer:

C — 3**Explanation**

Residual (spare) capacity = Arc capacity – Current flow = $8 - 5 = 3$. The residual capacity tells us how much additional flow can be pushed through this arc in the forward direction before it is saturated.

A transportation problem has 3 factories (supply nodes) and 5 warehouses (demand nodes). How many decision variables and constraints does the LP model have?

A

15 variables, 8 constraints

B

8 variables, 15 constraints

C

15 variables, 15 constraints

D

8 variables, 8 constraints

Correct Answer:

A — 15 variables, 8 constraints**Explanation**

Decision variables = sources $\times$ destinations = $3 \times 5 = 15$ (one per factory-warehouse route). Constraints = supply constraints + demand constraints = $3 + 5 = 8$. This is a standard result for balanced transportation LPs.

Three workers (P, Q, R) are assigned to three tasks (1, 2, 3). Time (hours):

P: Task1=4, Task2=9, Task3=3

Q: Task1=8, Task2=6, Task3=7

R: Task1=5, Task2=3, Task3=4

What assignment MINIMISES total time?

A

P→1, Q→2, R→3 (total = 14 h)

B

P→3, Q→2, R→1 (total = 14 h)

C

P→3, Q→1, R→2 (total = 14 h)

D

P→1, Q→3, R→2 (total = 14 h)

Correct Answer:

B — $P \rightarrow 3, Q \rightarrow 2, R \rightarrow 1$ (14 h)**Explanation**

Check key options:

- $P \rightarrow 3$ (3) + $Q \rightarrow 2$ (6) + $R \rightarrow 1$ (5) = 14 ✓
- $P \rightarrow 1$ (4) + $Q \rightarrow 2$ (6) + $R \rightarrow 3$ (4) = 14 ✓
- $P \rightarrow 3$ (3) + $Q \rightarrow 1$ (8) + $R \rightarrow 2$ (3) = 14 ✓

Multiple optimal solutions exist. The Hungarian method confirms minimum = 14 h.

Correct: $P \rightarrow 3, Q \rightarrow 2, R \rightarrow 1$.

A maximisation LP has two corner point optimal solutions at $(4, 6)$ and $(8, 2)$, both giving $\text{obj} = 40$. Which statement is TRUE?

A

The LP is infeasible

B

The LP is unbounded

C

Every point on the line segment between $(4,6)$ and $(8,2)$ is also optimal

D

 $(4,6)$ is the unique optimum because it has a higher x_2 value

Correct Answer:

C — Every point on the segment between the two corners is optimal**Explanation**

When two adjacent corner points yield the same objective value, the objective function is parallel to that edge. All convex combinations of the two corners (i.e., every point on the segment) are also optimal, giving infinitely many optimal solutions.

An LP has constraints $x_1 \geq 10$ and $x_1 \leq 6$ with $x_1 \geq 0$. What condition does this produce?

A

Multiple optimal solutions

B

An unbounded solution

C

Infeasibility — no feasible region exists

D

A degenerate solution

Correct Answer:

C — Infeasibility**Explanation**

The constraints $x_1 \geq 10$ and $x_1 \leq 6$ contradict each other — no value of x_1 can satisfy both simultaneously. This makes the feasible region empty, so the LP is infeasible (no solution exists).

In a maximisation LP, the feasible region extends infinitely in the direction of increasing objective value. What is this called?

A

Degeneracy

B

Infeasibility

C

Redundancy

D

Unboundedness

Correct Answer:

D — Unboundedness**Explanation**

If the feasible region is not bounded in the direction of optimisation (i.e., the objective can be improved indefinitely), the LP is unbounded. For minimisation, unboundedness means the objective can decrease without limit.

In a shortest-path IP, $x_{ij} = 1$ if arc $(i \rightarrow j)$ is used. For the SOURCE node s , the net flow constraint should be:

A $\Sigma(\text{out}) - \Sigma(\text{in}) = 0$

B $\Sigma(\text{out}) - \Sigma(\text{in}) = -1$

C $\Sigma(\text{out}) - \Sigma(\text{in}) = 1$

D $\Sigma(\text{in}) - \Sigma(\text{out}) = 1$

Correct Answer:

$$C - \Sigma(\text{out}) - \Sigma(\text{in}) = 1$$

Explanation

At the source, one unit of flow originates: net outflow = 1. So $\Sigma(\text{arcs out}) - \Sigma(\text{arcs in}) = 1$. Intermediate nodes have net flow = 0. The sink has net flow = -1 (one unit absorbed).

A sensitivity report shows a constraint with Shadow Price = 0 and Final Value = 45 against an RHS of 60. What does this tell you?

A

The constraint is binding and limits profit

B

The constraint is non-binding — there is 15 units of slack

C

The constraint should be removed from the model

D

Relaxing this constraint will increase profit by 15 units

Correct Answer:

B — Non-binding constraint with 15 units of slack**Explanation**

A shadow price of 0 indicates the constraint is non-binding (not active at the optimum). The slack = RHS – Final Value = $60 - 45 = 15$ units. Relaxing a non-binding constraint has no effect on the optimal objective value.

The Fundamental Theorem of Linear Programming states that if an optimal solution exists, it will always be found:

A

At the centroid of the feasible region

B

At a corner point (vertex) of the feasible region

C

On the objective function line

D

At the midpoint of a constraint boundary

Correct Answer:

B — At a corner point of the feasible region**Explanation**

For any LP with a bounded feasible region, if an optimal solution exists it occurs at a corner point (extreme point/vertex) of the feasible polytope. This underpins both the graphical method and the Simplex algorithm.

Q25

3 pts

MARKOV CHAINS

A Markov chain state is called 'absorbing' if:

A

It has the highest probability of being visited

B

Once entered, it is impossible to leave

C

It is visited most frequently in the long run

D

It has a self-loop probability of exactly 0.5

Correct Answer:

B — Once entered, it is impossible to leave**Explanation**

An absorbing state has a self-transition probability of 1 ($P(\text{stay}) = 1$, $P(\text{leave}) = 0$). Once the chain enters an absorbing state it stays there forever. Examples include 'bankruptcy' in business models or 'cured' in medical models.

Q26

3 pts

MAXIMAL FLOW

In a Max Flow network, the maximum flow from source to sink equals:

A The sum of all arc capacities in the network

B The minimum cut capacity across all source-sink cuts

C The number of arcs in the shortest path

D The total supply at the source node

Correct Answer:

B — The minimum cut capacity**Explanation**

The Max-Flow Min-Cut Theorem is a fundamental result: the maximum flow that can pass from source to sink equals the capacity of the minimum cut — the bottleneck set of arcs whose removal disconnects source from sink with minimum total capacity.

An LP relaxation of a minimisation IP gives optimal values $x_1 = 2.7$, $x_2 = 1.3$ with $\text{obj} = 18.4$. A manager rounds to $x_1 = 3$, $x_2 = 1$. Why might this rounded solution be suboptimal or infeasible?

A Rounding always gives the optimal integer solution

B Rounding up x_1 may violate a $\leq$ constraint, and the true optimal integer solution may not be near the LP relaxation optimum

C Rounding is only valid for maximisation problems

D The LP relaxation objective is always equal to the IP objective

Correct Answer:

B — Rounding may violate constraints or miss the true optimum**Explanation**

Naive rounding has two problems: (1) rounding up variables may push the solution outside the feasible region (violating $\leq$ constraints); (2) even when feasible, rounded solutions can be far from the true IP optimum. Branch and Bound is the rigorous alternative.

A factory LP's sensitivity report shows:

Machine hours: Shadow Price = 12, RHS = 200 h

Allowable Increase = 30 h, Allowable Decrease = 50 h

The factory can rent 40 extra machine hours at £9/hour. Should it do this?

A No — renting 40 h exceeds the allowable increase of 30 h, so the shadow price no longer applies

B Yes — profit gain = $40 \times 12 = £480$, which exceeds the cost of $40 \times 9 = £360$

C Yes — any extra capacity always improves profit

D No — the shadow price of 12 is less than the rental cost of 9

Correct Answer:

A — No: 40 h exceeds the allowable increase of 30 h**Explanation**

The shadow price of £12/h is only valid up to +30 h. Renting 40 h exceeds this range, so the shadow price no longer applies and we cannot use it to predict the profit gain. The manager would need to re-solve the LP for 240 h before deciding. Within the valid range (≤ 30 h), profit gain (£12/h) > rental cost (£9/h) so renting up to 30 h would be worthwhile.

Practice Complete

Review questions you found challenging and revisit those topics before your exam.

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